1. Mask format

- Each method outputs a **binary mask**: only 0 (not cloud) or 1 (cloud).
- The mask is the **same size as the tile**: same height and width, no cropping. If a method works on a shrunk image, resize the mask back to full size by copying pixel values (nearest neighbor).
- We use fixed **1024 x 1024 pixel tiles** at full resolution.
- Save one file per tile, with the tile's location info attached
- The method makes the final cloud / not cloud call itself, using a threshold chosen before scoring.

2. Uncertain pixels

Some pixels have no easy right answer. In the **reference mask** we mark them 255, which means "ignore." Ignored pixels are left out of the scoring for every method, so all methods are judged on the same pixels. Methods never output 255.

We mark a pixel as ignore when it is:

- **empty border** (no image data), or
- **unclear**: cloud edges, thin cloud, or haze where two people would label it differently.

Other labeling rules:

- Smoke, bright ground (sand, rooftops, snow), and cloud shadow are labeled 0 (not cloud). We tag smoke and bright ground so we can see how often each method confuses them with cloud.
- If more than 20% of a tile is ignored, flag that tile.

3. Metrics

We only count pixels that are not ignored. Each pixel is one of:

- **TP**: method said cloud, and it is cloud
- **FP**: method said cloud, but it is not
- **FN**: method said not cloud, but it is cloud
- **TN**: method said not cloud, and it is not
- **Precision** = TP / (TP + FP). Of what the method called cloud, how much was real? Low means false alarms.
- **Recall** = TP / (TP + FN). Of the real cloud, how much did the method find? Low means missed cloud.

- **F1** = 2 x Precision x Recall / (Precision + Recall). One score that balances precision and recall.
- **IoU** = TP / (TP + FP + FN). How much the predicted cloud overlaps the real cloud.
- **Main score:** add up TP, FP, FN across all test tiles, then compute the metrics once.
- **Also report:** the score for each tile, and the difference between predicted and real cloud cover (%).
- If a tile has no real cloud and the method predicts none, its IoU can't be computed, so skip it in the per-tile average and say how many were skipped.
- Also show results for the smoke and bright-ground tags, and for thin cloud.

4. Reference set

- We label the tiles by hand. Two people label a few of the same tiles, and how much they agree is the best score any method can fairly be expected to reach.
- Tiles should include clear, partly cloudy, and very cloudy scenes, plus smoke and bright-ground cases.
- Split tiles into **dev** (used to pick thresholds) and **test** (scored once at the end). Split by scene, so similar neighboring tiles don't end up in both.

5. Rules

1. Pick thresholds using dev tiles only.
2. Everyone uses the same scoring code from the repo.
3. Every results table lists the method version, thresholds, number of tiles, and ignored fraction.